VS-04 Estimation, Analysis, and Statistical Modeling

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1. Purpose and value created

VS-04 is where data becomes statistical information. It takes processed microdata and applies statistical methods, estimation techniques, and analytical models to produce results that describe a population: estimates, aggregates, indices, time series, and model outputs. VS-04 does not change microdata values. It computes new objects from them. Every result carries the method and parameters that produced it, a measure of its accuracy, and its lineage to the inputs. The value created is **analytical insight and statistical credibility**. Any figure that VS-04 produces can be regenerated and defended, and its fitness for purpose and coherence with related statistics have been checked.

VS-04 corresponds to the business domain **Estimation and closed-loop** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

- **Subject-matter statistician or analyst:** the stages in attribute 3, which describe what happens in each production cycle.
- **Methodologist or statistical designer:** the Design processes in attribute 4.
- **System builder or IT staff:** the Implementation processes in attribute 4.
- **Quality or compliance manager:** the Management processes in attribute 4.

- **Manager who wants the big picture:** attributes 1, 2, and 8. Examples and end-to-end scenarios are part of the SAF Toolkit.

> **i** Signature discipline: insight, transparently produced
>
> VS-04 guarantees methodological transparency, reproducibility, and statistical soundness. Every estimate carries the methodology and parameters that produced it. It carries an accuracy measure, such as a sampling error, a confidence interval, or a model diagnostic. It carries its lineage to the processed inputs, so any figure can be regenerated from State #3. Fitness for purpose and coherence with related and earlier statistics are explicit acceptance criteria.

> **i** Metadata inherited from VS-03
>
> VS-04 receives Processed Data together with the metadata VS-03 attached: the edit lineage, the definitions of derived variables, and the imputation flags. The imputation flags matter here. Estimation must reflect the extra uncertainty that imputed values introduce, for example through variance estimation that accounts for imputation. The edit and derivation lineage is carried through, so every estimate stays traceable to its observed and imputed inputs.

PLACEHOLDER: ArchiMate view VS-04 (a) value stream and capabilities

2. Position and state transition

VS-04 is the fourth primary value stream. It sits between editing (VS-03) and product composition (VS-05). It receives Processed Data when VS-03 publishes the *Processed Data Registered* event, and it hands over Statistics to VS-05.

VS-04 takes data from State #3 to State #4, with the status *analytical*. Before VS-04, the data is edited, complete, and enriched microdata. After VS-04, it is a registered, immutable version of statistics. The estimates are calculated, accuracy and quality indicators are produced, the lineage is documented, internal review is complete, and initial disclosure control has been applied. The results can potentially be released, but they are not yet released. The quality gate, the minimum metadata set, and the protection of State #4 are described in *State #4: Statistics* and are not repeated here.

State #4 has two gate moments. VS-04 passes the analytical gate, which confirms that the results are methodologically sound, accurate, coherent, and initially disclosure-controlled. VS-05 then passes the approval gate, which creates a new version with the status *approved*. Some processes start in VS-04 without passing VS-01 to VS-03 of their own. Macro statistics, such as national accounts, often start from Processed Data or Statistics that other domains have registered (see *Applying the steady states pragmatically*).

From	To	Gate	Event out
#3 Processed Data	#4 Statistics (status *analytical*)	Analytical gate of State #4 (methodological soundness, accuracy measures, coherence, model diagnostics, initial disclosure control, reproducibility, metadata and lineage)	*Output Data Registered*

PLACEHOLDER: ArchiMate view VS-04 (c) state handover and events: Processed Data (#3) in the upstream domain, the estimation service at the boundary (weighting, estimation, modeling, validation, initial disclosure control), Statistics (#4, analytical) inside the estimation domain

3. Stages

The stages describe the value that VS-04 adds in every production cycle. Together they form the execution process type of VS-04. In a mature steady state, the estimation strategy, the weighting method, and the models have been designed and built in advance (see attribute 4). Execution then focuses on running the cycle, computing weights, estimates, and models, and recording the methodology, parameters, and accuracy, so that every output can be reproduced and defended.

Stage	Name	Value added
VS-04.1	Confirm Estimation and Analytical Approach	The right, approved method is selected for this cycle's data and objective
VS-04.2	Apply Weighting and Calibration	Bias from sampling, non-response, and coverage is corrected
VS-04.3	Estimate and Aggregate Data	Representative statistics and aggregates are available, each with its accuracy
VS-04.4	Perform Model-Based Analysis	Relationships, trends, and small-domain estimates add insight beyond direct estimates
VS-04.5	Validate and Document Results	Sound, coherent, documented, and initially protected statistics are registered (State #4)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-04.1 Confirm Estimation and Analytical Approach

Value created: *The right, approved method selected for this cycle's data and objective.*

The stage starts when VS-03 publishes the *Processed Data Registered* event, or when a registered input from another domain is available for a downstream process. In a mature steady state, the methodology has been designed already. Execution in each cycle therefore confirms the approved approach for the data at hand, rather than choosing a new one. The team verifies that the calibration totals and reference sources are current. It confirms that the target parameters and domains of estimation still apply. A significant change, such as a new domain or a revision of the method, goes back to Design instead of being solved during the cycle.

Activity	Key outputs	SAF process	Other reference
Confirm or select the approved estimation and analytical approach for the cycle	Confirmed approach for the cycle	PRD-2.5 Design processing and analysis	1.3.5 Estimating population parameters
Verify that calibration totals and reference sources are current	Reference source confirmation	DLM-9 Manage reference and master data	CSDA Reference Data

Activity	Key outputs	SAF process	Other reference
Confirm the target parameters and domains of estimation	Confirmed estimation scope	PRD-5.6 Calculate weights	
Flag changes in methodology for design review	Methodology change exception (if any)	OVA-1 Quality Management	GAMSO Quality Management

VS-04.2 Apply Weighting and Calibration

Value created: *Bias from sampling, non-response, and coverage corrected, so that results represent the population.*

This stage computes the weights. The weighting engine applies the design weights and the non-response adjustment. The calibration engine then calibrates the weights to the confirmed population totals, using the designed auxiliary variables and enforcing the weight bounds. Convergence and the resulting distribution of weights are checked against the designed constraints. The calibration totals, auxiliary variables, and convergence diagnostics are recorded, so that the weighting can be reproduced. For register-based statistics without a sample, this stage may only apply coverage adjustments, or it may be skipped and the reason recorded.

Activity	Key outputs	SAF process	Other reference
Compute design weights and the non-response adjustment	Adjusted weights	PRD-5.6 Calculate weights	1.3.5 Estimating population parameters
Calibrate to the confirmed population totals with the designed auxiliary variables	Calibrated weights	PRD-5.6 Calculate weights	2.1.4 Statistical register management
Check convergence and the weight distribution against the bounds	Weight diagnostics	PRD-5.6 Calculate weights	
Record calibration totals, auxiliary variables, and diagnostics as lineage	Weighting lineage	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

VS-04.3 Estimate and Aggregate Data

Value created: *Representative statistics and aggregates, each carrying its accuracy.*

This stage computes the target estimates and aggregates. The estimation engine computes the target parameters with the calibrated weights. The aggregation pipeline computes the specified aggregates across the designed domains, with the agreed conventions for units and rounding. The variance estimation computes sampling errors and confidence intervals. It reflects both the survey design and the extra uncertainty from imputation, which the imputation flags of VS-03 signal. Every estimate is produced with its methodology, weights, and accuracy measure attached.

Activity	Key outputs	SAF process	Other reference
Compute the target estimates with the calibrated weights	Point estimates	PRD-5.7 Calculate aggregates	1.3.5 Estimating population parameters
Compute aggregates across the designed domains, with units and rounding	Output aggregates	PRD-5.7 Calculate aggregates	

Activity	Key outputs	SAF process	Other reference
Compute sampling errors and confidence intervals, including imputation uncertainty	Accuracy measures	PRD-6.1 Prepare draft outputs	2.1.3 Statistical quality management
Attach methodology, weights, and accuracy to every estimate	Estimate metadata	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

PLACEHOLDER: VS-04 estimation flow: processed microdata with imputation flags passing weighting and calibration against register totals, then estimation and aggregation per domain, with variance estimation producing an accuracy measure attached to every estimate

VS-04.4 Perform Model-Based Analysis

Value created: *Insight beyond direct estimates: relationships, trends, and estimates for small domains.*

This stage runs the model-based analyses specified in Design. Examples are regression models, time-series and forecasting models, and small-area models. The diagnostics of each model are computed and checked against the designed bounds. Seasonal adjustment is applied where it applies. Results are interpreted in context, and their assumptions and limitations are documented. As with direct estimates, every model output carries its specification, parameters, diagnostics, and uncertainty, so it can be reproduced and defended. Not every product needs this stage; where no models are specified, the stage is skipped.

Activity	Key outputs	SAF process	Other reference
Run the designed analytical, forecasting, and small-area models	Model outputs	PRD-6.1 Prepare draft outputs	1.4.1 Statistical (output) analysis
Compute model diagnostics and check them against the designed bounds	Model diagnostics	PRD-6.2 Validate outputs	
Apply seasonal adjustment or other time-series treatment where applicable	Adjusted series	PRD-6.1 Prepare draft outputs	
Interpret results and document assumptions and limitations	Analytical commentary	PRD-6.3 Interpret and explain outputs	

VS-04.5 Validate and Document Results

Value created: *Statistically sound, coherent, documented, and initially protected statistics, ready for handover.*

This stage validates the results and registers the official version. Results are checked for statistical soundness and consistency, and for coherence with earlier periods and related statistics. Significant movements are explained, and revisions are handled according to the revision policy. Initial statistical disclosure control is applied according to the designed policy. Disclosure risk is assessed, and protection methods such as suppression, rounding, or perturbation are applied, so that the output can potentially be released. The data must then pass the analytical gate of State #4. A persistent, immutable version is created with the status *analytical*, a unique identifier, and full lineage, and it is registered in the data catalog. Access permissions are applied. The stage ends by publishing the *Output Data Registered* event, which tells VS-05 that new validated statistics are available.

Output: Statistics (State #4, status *analytical*), the handover artifact to VS-05. Its structure and minimum metadata set are described in *State #4: Statistics*.

Activity	Key outputs	SAF process	Other reference
Validate soundness and coherence, explain movements, handle revisions	Validation report	PRD-6.2 Validate outputs; DLM-7 Revise data	2.1.3 Statistical quality management
Apply initial statistical disclosure control according to the designed policy	Disclosure-controlled outputs	PRD-6.4 Apply disclosure control	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
Document the methodology and the quality indicators	Methodology and quality documentation	PRD-6.5 Finalize outputs; OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Create an immutable, identified version with full lineage (status *analytical*)	Versioned Statistics	PRD-6.5 Finalize outputs; DLM-2 Promote data through a quality gate; DLM-5 Store and version data	CSDA Traceability
Register metadata and apply access permissions	Catalog registration, access-controlled dataset	DLM-3 Register and catalog data; DLM-4 Classify and protect data	SAF-P10 Security by design
Publish the *Output Data Registered* event to notify VS-05 Statistical Product Composition	Event notification	DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

PLACEHOLDER: VS-04 statistics version structure: the analytical statistics version linked to its methodology and parameters, weights, accuracy measures, model diagnostics, disclosure control record, and lineage to Processed Data (#3)

4. The four process types

VS-04 runs through four process types. They do not form a linear sequence but a governance cycle: each process type produces its own output, and that output feeds the others. Design produces metadata, Implementation produces the execution process, Execution produces the statistical data, and Management produces quality reports that feed back into Design. The generic pattern is described in the Overview of this chapter.

The sub-processes below are specific to VS-04. Their names describe the work in the context of estimation and analysis. Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. This keeps every VS-04 sub-process traceable to the reference process model in *Business Processes*.

Design: provides metadata

The design process type produces the **metadata artifacts** that govern all other process types. In VS-04 these are the estimation strategy, the weighting and calibration method, the aggregation and model specifications, and the validation and disclosure-control policy. Without them, estimation becomes ad hoc and cannot be defended, and management has no baseline to measure accuracy and coherence against. Design runs when a new product is introduced, when a method is revised, or when Management feeds back an improvement.

Design sub-process	Refines	Main outputs
Design Estimation Strategy and Accuracy Approach	PRD-2.5 Design processing and analysis; OVA-1 Quality Management	Estimation strategy, estimation specification, accuracy measurement plan, coherence criteria
Design Weighting and Calibration Methodology	PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	Weighting specification, calibration specification, weight constraint policy, register of calibration-total sources
Design Aggregation and Model Specifications	PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis	Aggregation specification, model specification, output-statistic metadata standard, time-series treatment
Design Validation and Disclosure-Control Policy	PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis; OVA-1 Quality Management	Validation policy, disclosure-control policy, output documentation standard, reproducibility design
Design Quality-Measurement and Versioning Framework	DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-1 Quality Management	Quality indicator set, versioning and revision policy, lineage specification, access control design

Design Estimation Strategy and Accuracy Approach

Refines: PRD-2.5 Design processing and analysis; OVA-1 Quality Management.

The starting point is the choice of how the statistics will be estimated. This sub-process selects the estimation strategy that fits the source and the analytical objective. Options are design-based estimation for sample surveys, aggregation for register-based statistics, and model-assisted or model-based estimation for small domains. It defines the target parameters, the reference populations, and the domains of estimation. It also defines how accuracy is measured: how sampling errors and confidence intervals are computed, and how the uncertainty introduced by imputation in VS-03 is reflected in variance estimation. Finally, the team defines the coherence criteria with related and earlier statistics.

Activity	Key outputs	SAF process	Other reference
Select the estimation strategy per source and objective: design-based, register-based, or model-based	Estimation strategy document	PRD-2.5 Design processing and analysis	1.3.5 Estimating population parameters
Define target parameters, reference populations, and domains of estimation	Estimation specification	PRD-2.5 Design processing and analysis	
Define how accuracy is measured, including the uncertainty from imputation	Accuracy measurement plan	PRD-2.5 Design processing and analysis	2.1.3 Statistical quality management
Define coherence criteria with related and earlier statistics	Coherence criteria	OVA-1 Quality Management	GAMSO Quality Management

Design Weighting and Calibration Methodology

Refines: PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data.

This sub-process designs how bias from sampling, non-response, and coverage is corrected through weighting. The team specifies the design weights and the model for non-response adjustment. It specifies the calibration

method, such as calibration to known population totals from registers. The auxiliary variables and the calibration totals are defined, together with their authoritative sources and versions. Bounds and acceptance constraints on weights are defined to prevent extreme weights. Recording the source and version of every calibration total makes the weighting reproducible later.

Activity	Key outputs	SAF process	Other reference
Specify design weights and the non-response adjustment	Weighting specification	PRD-2.5 Design processing and analysis	1.3.5 Estimating population parameters
Specify the calibration method, auxiliary variables, and calibration totals	Calibration specification	PRD-2.5 Design processing and analysis	
Define weight bounds and acceptance constraints	Weight constraint policy	PRD-2.5 Design processing and analysis	
Define the authoritative source and version of each calibration total	Register of calibration-total sources	DLM-9 Manage reference and master data	CSDA Reference Data; 2.1.4 Statistical register management

Design Aggregation and Model Specifications

Refines: PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis.

This sub-process specifies what is computed. For aggregation, each output statistic is defined by its formula, its domains and breakdowns, and its conventions for units and rounding. For analysis, the model-based methods are specified with their inputs, assumptions, and the diagnostics that must be produced and checked. Examples are regression, time-series and forecasting models, and small-area models. Where series are seasonally adjusted, the method and its parameters are specified. Each output statistic receives its metadata and classification, in line with SAF-P09.

Activity	Key outputs	SAF process	Other reference
Specify output aggregates: formulas, domains and breakdowns, units, rounding	Aggregation specification	PRD-2.1 Design outputs and dissemination	
Specify analytical and forecasting models: inputs, assumptions, diagnostics	Model specification	PRD-2.5 Design processing and analysis	1.4.1 Statistical (output) analysis
Define metadata and classification for every output statistic	Output-statistic metadata standard	PRD-2.1 Design outputs and dissemination	SAF-P09 No data without metadata and classification
Define seasonal adjustment and other time-series treatment where applicable	Time-series treatment specification	PRD-2.5 Design processing and analysis	

Design Validation and Disclosure-Control Policy

Refines: PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis; OVA-1 Quality Management.

This sub-process designs how results are validated and protected before handover. The validation criteria cover statistical soundness: plausibility, consistency, and model diagnostics within bounds. They also cover coherence with related and earlier statistics, and the revision policy. The policy for statistical disclosure

control is designed. It sets the criteria for disclosure risk of tables and output microdata, and the protection methods, such as suppression, rounding, and perturbation. Reproducibility is designed in: an output must be regenerable from the processed input and the recorded methodology and parameters. Together, these rules make up the VS-04 part of the analytical gate of State #4.

Activity	Key outputs	SAF process	Other reference
Define validation criteria: soundness, coherence, model diagnostics, revision policy	Validation policy	PRD-2.5 Design processing and analysis; OVA-1 Quality Management	GAMSO Quality Management
Define disclosure-risk criteria and protection methods	Disclosure-control policy	PRD-2.1 Design outputs and dissemination	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
Define the requirements for output metadata and methodology documentation	Output documentation standard	PRD-2.1 Design outputs and dissemination	SAF-P09 No data without metadata and classification
Design reproducibility: regenerate outputs from inputs, methodology, and parameters	Reproducibility design	PRD-2.5 Design processing and analysis	SAF-P10 Security by design

Design Quality-Measurement and Versioning Framework

Refines: DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-1 Quality Management.

This last design sub-process makes sure that accuracy is measured and that outputs can be traced. It defines the quality indicators to produce, such as the sampling error, the coefficient of variation, response and coverage rates, and model fit. It defines the versioning and revision policy: how versions are identified, what triggers a revision, and how immutability and retention are handled. It defines the lineage from each statistic back to its processed inputs and the method applied. Role-based access controls for outputs are designed in line with SAF-P10. Unreleased statistics are confidential until release, so access is limited to the roles that need them.

Activity	Key outputs	SAF process	Other reference
Define quality indicators: sampling error, coefficient of variation, response and coverage, model fit	Quality indicator set	OVA-1 Quality Management	GAMSO Quality Management
Define the versioning and revision policy for outputs	Versioning and revision policy	DLM-1 Plan and design data assets	CSDA Traceability
Define the lineage from each statistic to its processed inputs and method	Lineage specification	DLM-1 Plan and design data assets	CSDA Traceability
Design role-based access control for unreleased outputs	Access control design	DLM-4 Classify and protect data	SAF-P10 Security by design; 3.3.4 Logical identity and access control

PLACEHOLDER: VS-04 design process decomposition: inputs (State #3 metadata, methodology standards, register totals, legal and confidentiality requirements), the five design sub-processes in order, and the design artifact each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

The implementation process type takes the metadata artifacts from Design and **builds, configures, and tests** the estimation and analysis pipeline until it is ready for production. Nothing runs operationally until this process type is complete. Components are reused where possible and built only where needed, in line with SAF-P01. Calibration, variance-estimation, and seasonal-adjustment tools shared in the NSI community are good candidates for reuse.

Implementation sub-process	Refines	Main outputs
Build Weighting and Calibration Engines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data	Weighting engine, calibration engine, connection to calibration totals
Build Estimation and Aggregation Pipelines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Estimation pipeline, variance component, aggregation pipeline
Build Modeling Environment, Validation, and SDC Tooling	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data	Modeling environment, validation tooling, disclosure-control tooling, versioning store, events
Test and Validate the Pipeline	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems	Test reports, reproducibility and benchmark reports, production-ready pipeline

Build Weighting and Calibration Engines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data.

This sub-process turns the weighting and calibration designs into operational tools. The weighting engine is built or configured to compute design weights and non-response adjustments. The calibration engine is built or configured to calibrate to the designed totals with the specified auxiliary variables, enforcing the weight bounds. The connections to calibration totals are configured to the authoritative register and reference sources, with the version of each total fixed and recorded. All weighting logic is version-controlled in line with SAF-P10. Any set of weights can therefore be reproduced from its inputs and the recorded versions.

Activity	Key outputs	SAF process	Other reference
Build or configure the weighting engine: design weights, non-response adjustment	Configured weighting engine	PRD-3.2 Reuse or build processing and analysis components	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes

Activity	Key outputs	SAF process	Other reference
Build or configure the calibration engine with auxiliary variables and bounds	Configured calibration engine	PRD-3.2 Reuse or build processing and analysis components	1.3.5 Estimating population parameters
Configure connections to versioned register sources for calibration totals	Integration of calibration totals	DLM-9 Manage reference and master data	CSDA Reference Data
Put all weighting and calibration logic under version control	Versioned methodology repository	PRD-3.4 Configure workflows	SAF-P10 Security by design

Build Estimation and Aggregation Pipelines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

In this sub-process the designed estimation and aggregation specifications become automated services. The estimation engine computes the target parameters with the designed accuracy measures. The variance component computes sampling errors and confidence intervals. It accounts for the survey design and for the uncertainty from imputation, using the imputation flags of VS-03. The aggregation pipeline computes the specified aggregates across the designed domains, with the agreed units and rounding. Every output is produced with its methodology, parameters, weights, and accuracy measure attached.

Activity	Key outputs	SAF process	Other reference
Build the estimation engine for the designed target parameters	Estimation pipeline	PRD-3.2 Reuse or build processing and analysis components	1.3.5 Estimating population parameters
Build the variance component: design and imputation uncertainty	Variance and accuracy component	PRD-3.2 Reuse or build processing and analysis components	2.1.3 Statistical quality management
Build the aggregation pipeline: domains, units, rounding	Aggregation pipeline	PRD-3.2 Reuse or build processing and analysis components	
Attach methodology, weights, and accuracy measure to every output	Binding of output metadata	PRD-3.4 Configure workflows	SAF-P09 No data without metadata and classification

Build Modeling Environment, Validation, and SDC Tooling

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data.

This sub-process implements model-based analysis, validation, and disclosure control. The modeling environment is built or configured to run the specified models with their diagnostics, and seasonal adjustment where it applies. Validation tooling implements the designed checks on soundness and coherence, including comparison with earlier periods and related statistics. The tooling for statistical disclosure control implements the designed risk checks and protection methods. The versioning store is built so that each accepted output version is immutable, identified, reproducible, and accompanied by its lineage. Event notifications are configured, so that VS-05 is alerted when a new version of Statistics is registered.

Activity	Key outputs	SAF process	Other reference
Build or configure the modeling environment: models, diagnostics, seasonal adjustment	Modeling environment	PRD-3.2 Reuse or build processing and analysis components	1.4.1 Statistical (output) analysis

Activity	Key outputs	SAF process	Other reference
Implement validation tooling: soundness, coherence, comparison with earlier periods	Validation tooling	PRD-3.2 Reuse or build processing and analysis components	
Implement disclosure-control tooling: risk checks, suppression, rounding, perturbation	Disclosure-control tooling	PRD-3.2 Reuse or build processing and analysis components	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
Build the versioning store: immutable, identified, reproducible	Versioning store	DLM-5 Store and version data	CSDA Traceability
Configure the *Output Data Registered* event for downstream consumers	Event publication	PRD-3.4 Configure workflows; DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

Test and Validate the Pipeline

Refines: PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems.

Before the pipeline goes into production, it must be thoroughly verified. Technical testing covers all individual components: weighting, calibration, estimation, variance, aggregation, modeling, validation, disclosure control, and the versioning store. Integration testing checks the end-to-end flow from the *Processed Data Registered* event to the registration of Statistics. A reproducibility test confirms that an output can be regenerated identically from its input and the recorded methodology and parameters. A benchmark test compares results with a known benchmark or with an earlier series. Accuracy measures and model diagnostics are checked against expectations. The sub-process ends with putting the production system into use and documenting the operational procedures.

Activity	Key outputs	SAF process	Other reference
Test all individual components	Unit test reports	PRD-3.5 Test production systems	
Test integration: end-to-end flow from processed intake to versioned registration	Integration test report	PRD-3.5 Test production systems	
Run the reproducibility test: regenerate an output from inputs and methodology	Reproducibility report	PRD-3.6 Test statistical business process	CSDA Traceability
Run the benchmark test against known or earlier results	Benchmark test report	PRD-3.6 Test statistical business process	
Check accuracy measures and model diagnostics against expectations	Accuracy verification report	PRD-3.6 Test statistical business process	2.1.3 Statistical quality management
Finalize the production system and document operational procedures	Operational procedures manual, production-ready pipeline	PRD-3.7 Finalize production systems	

PLACEHOLDER: VS-04 implementation pipeline: the five design artifacts feeding the three build sub-processes, followed by technical testing, reproducibility testing, and benchmark testing, resulting in a production-ready estimation pipeline and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-04 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts when VS-03 publishes the *Processed Data Registered* event, when a registered input from another domain is available, or when a revision is due. The stages run in order; VS-04.4 is skipped where no models are specified. Execution applies the designs and the solution as they are; a new domain or a method that no longer fits goes back to Design. Every run ends in one of three outcomes: registered as *analytical*, registered with documented quality restrictions, or returned for re-estimation.

Aspect	VS-04
Trigger	*Processed Data Registered*; a registered input from another domain; a scheduled revision
Cadence	Per production cycle, in line with the release calendar
Performed by	ROL-15 Subject-matter statistician, ROL-07 Methodologist, ROL-03 Data steward
Input	Processed Data quality version (#3) or other registered input, calibration totals, approved methodology
Output	Registered version of Statistics (#4, *analytical*), its methodology and accuracy documentation
Control point	Analytical gate of State #4 in VS-04.5 Validate and Document Results
Exception path	Implausible results back to VS-04.2 Apply Weighting and Calibration or VS-04.3 Estimate and Aggregate Data; a data problem back to VS-03 Data Editing and Enrichment; a method problem back to Design; disclosure risk above the threshold escalated to ROL-16 Disclosure control specialist

Management: provides quality reports

The management process type runs alongside and after execution. It monitors the performance of the estimation pipeline, assesses the accuracy and coherence of the outputs, makes sure that every result can be reproduced and is disclosure-safe, and feeds improvements back into Design. This closes the governance cycle.

Management sub-process	Refines	Main outputs
Monitor Process Performance	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard, throughput report, paradata records
Assess Output Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Accuracy report, coherence report, revision analysis, trend analysis
Manage Methodology and Output-Version Lifecycle	OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-8 Monitor data quality, lineage and use	Methodology version register, metadata quality report, version store audit

Management sub-process	Refines	Main outputs
Audit and Compliance Reporting	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Reproducibility audit, disclosure-control and methodology compliance reports to VS-07 Governance and Compliance
Continuous Improvement	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, improvement tracking log

Monitor Process Performance

Refines: OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs.

The operational performance of VS-04 must be tracked systematically across production cycles. The process owner and the quality manager collect and aggregate the key performance indicators (see attribute 9). They monitor throughput against the release calendar, because VS-04 is often on the critical path to publication. They track paradata: which methods, parameter versions, and software versions were run, when, by whom, and how many manual interventions took place. This information feeds dashboards that give the process owner a current view of the health of the domain. The approach follows the Quality Management capability of GAMSO.

Activity	Key outputs	SAF process	Other reference
Collect and aggregate the key performance indicators per production cycle	Process performance dashboard	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	GAMSO Quality Management
Monitor throughput against the release calendar	Throughput report	OVA-4 Process Data Management	
Track paradata: methods, parameter and software versions, interventions	Paradata records	OVA-4 Process Data Management	
Produce process performance dashboards	Dashboard for the process owner	OVA-4 Process Data Management	

Assess Output Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality of the outputs is evaluated against the criteria from *Design Validation and Disclosure-Control Policy* and *Design Quality-Measurement and Versioning Framework*. Accuracy covers sampling error, the coefficient of variation, and model fit. Coherence covers consistency with related and earlier statistics. Timeliness compares production with the release calendar. Reliability is assessed through revision analysis, which compares preliminary and final figures. Quality indicators are produced per statistic, per output, and per cycle. Trends are tracked to detect falling accuracy, growing revisions, or models that need recalibration. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Evaluate accuracy: sampling error, coefficient of variation, model fit	Accuracy report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess coherence with related and earlier statistics	Coherence report	OVA-1 Quality Management	CSDA Data Quality

Activity	Key outputs	SAF process	Other reference
Carry out revision analysis: preliminary against final figures	Revision analysis	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management
Identify quality trends across statistics and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	CSDA Data Quality

Manage Methodology and Output-Version Lifecycle

Refines: OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-8 Monitor data quality, lineage and use.

The estimation methods and the output versions they produce must stay current, complete, and well governed. This sub-process governs the life cycle of estimation methods, weighting and calibration specifications, and models. It versions them, manages transitions, and retires methods that are no longer used. A change of method often has consequences for earlier periods, so the need to revise back series is assessed with every change. The sub-process verifies that every output carries its full methodology, weights, accuracy measures, and lineage, as a compliance check on SAF-P09. It also governs the versioning store: retention, immutability, revisions, and the reconciliation of catalog entries with stored versions.

Activity	Key outputs	SAF process	Other reference
Govern the life cycle and versions of estimation methods, weighting, and models	Methodology version register	OVA-2 Metadata Management	SAF-P03 Active life cycle management
Assess the impact of method changes on back series	Back-series impact assessment	DLM-7 Revise data	
Verify metadata completeness: methodology, weights, accuracy, lineage	Metadata quality report	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Govern the versioning store: retention, immutability, revisions, reconciliation	Version store audit	DLM-5 Store and version data; DLM-7 Revise data	CSDA Traceability
Track metadata lineage from processed data to statistics	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Traceability

Audit and Compliance Reporting

Refines: OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use.

This sub-process produces the evidence that VS-04 operates within methodological, legal, and organizational requirements. Reproducibility audit reports show that every figure can be regenerated from the processed input and the recorded methodology and parameters. This supports methodological transparency and accountability. Disclosure-control reporting confirms that disclosure control was applied and that outputs meet the risk criteria, in line with SAF-P11. Methodology compliance reporting confirms that approved estimation and analytical methods were applied. Security compliance reporting verifies that the designed access controls and the tamper-evident record of methods were applied. These reports meet the obligation of SAF capability 4.1.5 *Accountability* and feed VS-07.5 *Audit, Certify and Report Compliance*.

Activity	Key outputs	SAF process	Other reference
Produce reproducibility audit reports: every figure can be regenerated	Reproducibility audit summary	OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	4.1.5 Accountability
Report on the disclosure control applied and its outcomes	Disclosure-control report	OVA-1 Quality Management	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
Report on methodology compliance: approved methods applied	Methodology compliance report	OVA-1 Quality Management	4.1.5 Accountability
Report on security compliance: access controls, integrity of the methodology record	Security compliance report (per cycle)	OVA-1 Quality Management	SAF-P10 Security by design; 3.3.3 Security incident management

Continuous Improvement

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This last management sub-process closes the governance cycle by analyzing performance and quality data and driving improvements. Trends across production cycles are examined: are sampling errors falling, are revisions getting smaller, is coherence improving, and do the models still hold? Improvement opportunities are identified. Examples are refining the calibration to reduce variance, recalibrating or replacing models, improving the non-response adjustment, adopting better variance estimators, and shortening the time to publication. Recommendations are fed back to the design sub-processes, prioritized by impact and feasibility, and tracked through the following cycles. Problems that originate in the data are reported to VS-03. Where a new method is needed, the recommendation goes to VS-08. This is SAF capability 4.1.4 *Improvement management* in practice.

Activity	Key outputs	SAF process	Other reference
Analyze quality trends across production cycles	Improvement recommendations report	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Identify improvement opportunities: calibration, models, variance estimators, timeliness	Updated design inputs	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Feed recommendations back to the design sub-processes, to VS-03 Data Editing and Enrichment, or to VS-08 Innovation and Methodological Development	Design feedback (closing the cycle)	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Prioritize improvements by impact and feasibility	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management
Track the implementation of accepted improvements	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management

PLACEHOLDER: VS-04 management feedback loop: outputs and paradata from Execution flowing through the five management sub-processes, with dashboards to the process owner, accuracy and coherence reports to the quality manager, compliance reports to VS-07, data issues to VS-03, and improvement recommendations back to Design

Metadata and paradata per process type

In VS-04, metadata is what makes a published figure defensible: every estimate must carry the method, parameters, weights, and accuracy that produced it. The table shows what each process type contributes. Data metadata describes the outputs. Process metadata, also called paradata, describes how the process itself ran: who, when, and with which tools.

Process type	Data metadata produced	Process metadata (paradata) produced
Design	Estimation and weighting methodology, calibration totals and their sources, aggregation specifications, model specifications, validation and disclosure-control policy	Log of design decisions, methodology version, review and approval records
Implementation	Engine configurations, weighting and estimation parameters, model parameters, variance settings, disclosure-control parameters	Build log, test results, reproducibility test outcomes, deployment approvals
Execution	Methodology per estimate, weights, accuracy measures, model diagnostics, reference population, record of disclosure control applied	Operator identity, timing, software versions, parameter values, run identifiers
Management	Accuracy and coherence assessments, revision analyses, lineage assessments, trend analyses	Accuracy dashboards, coherence reports, improvement decisions, audit findings

PLACEHOLDER: VS-04 business object lifecycle: design artifacts govern implementation assets, which produce the pipeline that turns Processed Data (#3) into Statistics (#4, analytical), generating methodology, accuracy, and audit records that management assesses and feeds back to design

5. Actors and roles

VS-04 is an internal value stream driven by methodology and subject-matter expertise. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the estimation methodologist is ROL-07 Methodologist, the model designer is ROL-06 Statistical designer, and the survey statistician and the subject-matter reviewer are ROL-15 Subject-matter statistician. VS-03, which supplies the processed data, is a neighboring value stream and not an actor; the relation is governed by the handover interface in attribute 8. **[VC20]**

Actor or role	ID	Process type	Responsibility in VS-04 Estimation, Analysis, and Statistical Modeling
International statistical organization	ACT-07	External	Issues methodological guidance and regulations on estimation, seasonal adjustment, and accuracy reporting (for example Eurostat)
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-04 Estimation, Analysis, and Statistical Modeling
Data owner	ROL-02	Management	Accountable for analytical versions of State #4; decides on internal access and reuse
Data steward	ROL-03	Execution	Registers output versions, maintains catalog entries, applies access controls, resolves gate exceptions
Data custodian	ROL-04	Execution	Operates storage, access control, and retention of unreleased statistics
Statistical designer	ROL-06	Design	Specifies output aggregates and analytical and forecasting models, with their inputs and diagnostics
Methodologist	ROL-07	Design, Execution	Designs the estimation strategy, weighting and calibration, and accuracy measurement; confirms the approach per cycle
Metadata and data architect	ROL-08	Design	Designs the metadata model for methodology, accuracy, and lineage of outputs
Data engineer	ROL-10	Implementation	Builds and configures the weighting, estimation, aggregation, and modeling components
Platform engineer	ROL-11	Implementation	Configures estimation services, connections to calibration-total sources, and access controls
Test manager	ROL-12	Implementation	Plans and carries out technical, reproducibility, and benchmark testing

Actor or role	ID	Process type	Responsibility in VS-04 Estimation, Analysis, and Statistical Modeling
Subject-matter statistician	ROL-15	Execution	Computes estimates, runs models, reviews results for plausibility and coherence, and signs off
Disclosure control specialist	ROL-16	Design, Execution	Designs the disclosure-control policy and assesses the disclosure risk of outputs
Quality manager	ROL-18	Management	Monitors accuracy and coherence indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures reproducibility and compliance with disclosure-risk rules, and produces audit reports

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-04 relies on, and what role they play. NSIs can use it to map the value stream to their own organization.

Capability group	Capabilities	Role in VS-04 Estimation, Analysis, and Statistical Modeling
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the estimation and weighting methods and the models, and build the estimation pipeline
1.3 Processing	1.3.5 Estimating population parameters, 1.3.7 Statistical disclosure	Compute weights, estimates, and aggregates, and apply initial disclosure control **[VC21]**
1.4 Analysis	1.4.1 Statistical (output) analysis	Run model-based analysis, validate, and interpret results **[VC21]**
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage output versions and methodology metadata, assess accuracy and coherence, use registers for calibration totals
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain weighting, estimation, aggregation, and modeling components
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.4 Logical identity and access control	Assess the disclosure risk of outputs and protect unreleased statistics

Capability group	Capabilities	Role in VS-04 Estimation, Analysis, and Statistical Modeling
4.1 Strategy and leadership	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, steward the estimation methods, plan improvements, report on reproducibility and compliance

Stages, processes, and functions

The table shows which SAF business processes realize each stage of VS-04 and which business functions they mainly use. It follows the mapping in *Business Processes, Mapping to value stream stages*. The activity tables in attribute 3 can name additional supporting processes. **[VC25]**

Stage	Realized by processes	Main functions
VS-04.1 Confirm Estimation and Analytical Approach	PRD-2.4 Design frame and sample; PRD-2.5 Design processing and analysis	FUN-3.1 Design and build statistical products; FUN-4.1 Manage statistical and analytical methodology
VS-04.2 Apply Weighting and Calibration	PRD-5.6 Calculate weights	FUN-3.4 Estimate and analyze
VS-04.3 Estimate and Aggregate Data	PRD-5.7 Calculate aggregates	FUN-3.4 Estimate and analyze
VS-04.4 Perform Model-Based Analysis	PRD-6.1 Prepare draft outputs; PRD-6.3 Interpret and explain outputs	FUN-3.4 Estimate and analyze
VS-04.5 Validate and Document Results	PRD-6.2 Validate outputs; PRD-6.4 Apply disclosure control; OVA-2 Metadata Management	FUN-3.4 Estimate and analyze; FUN-3.5 Protect statistical confidentiality; FUN-7.2 Manage metadata

PLACEHOLDER: ArchiMate view VS-04 (b) stages realized by processes and functions

Framework mappings

This reference table maps each process type of VS-04 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-2.1, PRD-2.5, DLM-1, DLM-4, DLM-9, OVA-1	Phase 2 Design (2.1, 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management, Data Quality	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.3 Statistical quality management

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Implementation	PRD-3.2, PRD-3.4 to PRD-3.7, DLM-3, DLM-5, DLM-9	Phase 3 Build (3.2, 3.4 to 3.7)	Capability Management	Data Analysis (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-5.6, PRD-5.7, PRD-6.1 to PRD-6.5, DLM-2 to DLM-5, DLM-7, DLM-9, OVA-1, OVA-2	Phase 5 Process (5.6 Calculate weights, 5.7 Calculate aggregates); Phase 6 Analyze (6.1 to 6.5, with 6.4 as initial disclosure control)	Production (execution)	Data Analysis (execution)	1.3.5 Estimating population parameters, 1.3.7 Statistical disclosure, 1.4.1 Statistical (output) analysis
Management	OVA-1, OVA-2, OVA-4, DLM-5, DLM-7, DLM-8, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-04 most directly.

Principle	Statement	Application in VS-04 Estimation, Analysis, and Statistical Modeling
SAF-P01	Reuse before open source, open source before buying, buying before building	Estimation, calibration, variance, and seasonal-adjustment tools are reused across products first. Where new tooling is needed, open-source solutions developed with other NSIs come before purchase or own development. **[VC3]**
SAF-P07	Implement loosely coupled processes and systems	Estimation is independent from editing and composition. Statistics (State #4) is the decoupling interface: downstream consumers use validated statistics without knowing the internals of estimation.

Principle	Statement	Application in VS-04 Estimation, Analysis, and Statistical Modeling
SAF-P08	Data is shared property	Statistics and output microdata, once validated, are available to all authorized internal consumers. Estimation is not repeated per product when the same validated output can serve several purposes, for example as input for macro statistics.
SAF-P09	No data without metadata and classification	Every estimate carries metadata: the methodology and parameters, the weights, the accuracy measures, the reference population, and the classification of derived statistics. This is what makes results transparent and reproducible.
SAF-P10	Security by design	Access controls, integrity verification, and a tamper-evident record of methods and parameters are built into the pipeline, so every figure can be attributed and reproduced. Unreleased statistics are protected against early disclosure.
SAF-P11	Privacy by design	Initial statistical disclosure control is applied to outputs, and the disclosure risk of estimates and output microdata is assessed by design, so the results can potentially be released.

i Principles that apply across the process types

Several principles apply to all four process types. SAF-P09 is the backbone of methodological transparency: Design defines the methodology and the required output metadata, Execution records method, parameters, weights, and accuracy for every estimate, and Management assesses output quality and coherence. SAF-P10 governs the tamper-evident record of methods and the reproducibility of outputs across all stages. SAF-P11 applies in particular to *Design Validation and Disclosure-Control Policy* and to VS-04.5, where disclosure risk is assessed before handover. SAF-P07 governs both the intake of State #3 from VS-03 and the handover of State #4 to VS-05.

8. Interfaces, services, and events

Boundary interfaces

VS-04 is loosely coupled (SAF-P07) to its neighbors through two boundary interfaces. Specifying each boundary explicitly makes the loose-coupling contract concrete and testable: the handover artifact, the metadata it must carry, the event that triggers it, and the point where responsibility transfers. Each interface exchanges a registered, immutable artifact plus its metadata, never internal working data. Neither side therefore needs knowledge of the other's internal processing.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-03 Data Editing and Enrichment → VS-04 Estimation, Analysis, and Statistical Modeling (intake)	Registered Processed Data quality version (#3)	Minimum metadata set of State #3, including edit lineage, derived-variable definitions, imputation flags and methods, quality assessment, and quality version identifier	*Processed Data Registered*	VS-03 Data Editing and Enrichment keeps custody of the quality version; VS-04 Estimation, Analysis, and Statistical Modeling takes responsibility for weighting, estimation, modeling, and validation on receipt of the event
VS-04 Estimation, Analysis, and Statistical Modeling → VS-05 Statistical Product Composition (handover)	Registered version of Statistics (#4, *analytical*)	Methodology and parameters, weights, accuracy measures, model diagnostics, record of disclosure control applied, quality indicators, version identifier, lineage to the processed source, classification	*Output Data Registered*	VS-04 Estimation, Analysis, and Statistical Modeling keeps custody of the analytical version; VS-05 Statistical Product Composition takes responsibility for composition and approval on receipt of the event

Business services

Service	Offered to	Realized in	Interfaces
Weighting and calibration service	Estimation teams	VS-04.2 Apply Weighting and Calibration	Weighting and calibration engines, connection to calibration totals
Estimation and aggregation service	Estimation teams, analysts	VS-04.3 Estimate and Aggregate Data	Estimation interface that returns estimates with accuracy measures; query interface for methodology and parameters
Modeling and analysis service	Analysts, methodologists	VS-04.4 Perform Model-Based Analysis	Modeling environment, diagnostics reports
Validation and disclosure-control service	Quality managers, disclosure control specialists	VS-04.5 Validate and Document Results	Validation and disclosure-control reports
Statistics provisioning service	VS-05 Statistical Product Composition, other internal products using the statistics as input	VS-04.5 Validate and Document Results	Event notification (*Output Data Registered*) and a catalog query interface to find output versions, their accuracy, and methodology

Service	Offered to	Realized in	Interfaces
Output version catalog service	All data consumers in the NSI	Management sub-process *Manage Methodology and Output-Version Lifecycle*	Search and browse interface for output versions, accuracy measures, methodology documentation, and lineage
Design and methodology service	Methodologists, statistical designers	All Design sub-processes of VS-04 Estimation, Analysis, and Statistical Modeling	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artifacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Processed Data Registered	Start of VS-04.1 Confirm Estimation and Analytical Approach for that version (published by VS-03 Data Editing and Enrichment)
Calibration totals updated	Review of calibration in VS-04.1 Confirm Estimation and Analytical Approach before the next run
Results validated / *Results rejected*	Outcome of validation in VS-04.5 Validate and Document Results; rejection starts re-estimation or a return to VS-03 Data Editing and Enrichment
Output Data Registered	Completion of VS-04.5 Validate and Document Results; triggers VS-05 Statistical Product Composition
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-04 event choreography between VS-03, the estimation teams, subject-matter statisticians, and process management across Design, Implementation, Execution (confirm approach, weight and calibrate, estimate, model, validate and register), and Management (key performance indicators, accuracy and coherence, compliance, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-04	Receives from VS-04
VS-07 Governance and Compliance	Methodology and confidentiality policy, approval of methods	Reproducibility audits, methodology and disclosure-control compliance evidence
VS-08 Innovation and Methodological Development	New estimation, modeling, and seasonal-adjustment methods	Methods that no longer fit, improvement needs
VS-09 Infrastructure and Platform Enablement	Estimation and modeling environments, versioning store, catalog, event interfaces	Capacity requirements, in particular around release dates
VS-10 Customer and Stakeholder Engagement	Requirements of users for accuracy, breakdowns, and timeliness	Information on accuracy and on methods applied

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
On-time output rate	Share of outputs produced within the window of the release calendar	Per output, per cycle
Average sampling error or coefficient of variation	Typical accuracy of key estimates	Per statistic, per cycle
Coherence check pass rate	Share of outputs that pass the coherence checks on the first run	Per cycle
Revision magnitude	Size of revisions against previously published figures	Per statistic, per cycle
Reproducibility rate	Share of outputs that can be regenerated identically from inputs and methodology	Per cycle
Processing time	Time from the *Processed Data Registered* event to the registration of #4	Per output, per cycle

10. Toolkit

Examples from NSI practice, the maturity checklist, and end-to-end scenarios for VS-04 are part of the SAF Toolkit.

Toolkit: SAF-TK-V04 VS-04 Estimation, Analysis, and Statistical Modeling companion [**VC17**]